

SAMADRITA ROY

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PROFESSIONAL SUMMARY

Frontend Engineer with 4+ years of experience designing, developing and maintaining scalable, responsive web applications using React.js, TypeScript and JavaScript (ES6+) across aviation, fintech, education and government domains. Strong hands-on expertise in React component-based architecture, GraphQL and REST API integration, GitHub-based version control, and CI/CD pipelines, combined with solid grounding in Agile/SAFe delivery, unit testing (Jest), code reviews, and SDLC best practices. Delivered measurable impact through ~70% faster development cycles via reusable component design, ~20% performance improvement, and ~90% WCAG 2.0/2.1 accessibility audit scores on production systems serving 750K+ users. Working knowledge of AI-assisted development tools and responsible AI usage in modern software engineering.

TECHNICAL SKILLS

Languages & Frameworks: JavaScript (ES6+), TypeScript, React.js, React Native, HTML5, CSS3/SCSS

State Management & APIs: Redux, GraphQL, REST APIs

Version Control & Tools: GitHub, Git, Visual Studio Code, Jira, Confluence, Webpack, Storybook, CI/CD

Testing & Quality: Jest, React Testing Library, Unit Testing, Peer Code Reviews

Architecture: Component-Based Architecture, Modular & Scalable UI Design, Non-Functional Requirements (NFR)

Methodology: Agile/Scrum, SAFe, Software Development Life Cycle (SDLC)

Accessibility & Performance: WCAG 2.0/2.1, Cross-Browser Compatibility, Lazy Loading, Render Optimization, Responsive Design

Emerging Technologies: Generative AI, Prompt Engineering, LLM Integration, AI-Assisted Development Tools

PROFESSIONAL EXPERIENCE

Frontend Developer (Custom Software Engineer track)

Feb 2022 – Present

Tata Consultancy Services — Pune, India

- Designed, developed and maintained 5+ enterprise-scale React.js and TypeScript applications supporting 750K+ users across aviation, fintech, education and government platforms, contributing independently as a subject matter expert (SME) across the SDLC.
- Built reusable, component-based React architecture managed through GitHub version control and CI/CD workflows, accelerating delivery speed by ~70% and improving code maintainability.
- Integrated GraphQL and REST APIs across 20+ production modules, ensuring efficient data flow and predictable state management with Redux.
- Improved application performance by ~20% through lazy loading, render optimization and efficient API consumption; achieved ~90% WCAG 2.0/2.1 accessibility audit scores.
- Collaborated with cross-functional teams of 30+ backend engineers, QA, designers and product stakeholders in Agile/Scrum (SAFe) ceremonies, sprint planning, unit testing and code reviews to deliver high-quality releases.
- Utilized AI-assisted development tools to improve productivity and code quality, following enterprise coding standards and responsible AI usage guidelines.

KEY PROJECTS

DSP Ireland – Government Pension Platform

Nov 2024 – Present

React, TypeScript, GraphQL, React Native, Strapi CMS

- Delivered enrollment and opt-in/opt-out workflows for a government pension platform serving 750K+ users, working with 60–70+ GraphQL API endpoints for structured, efficient data fetching.
- Built a custom BlocksRenderer component enabling dynamic, multilingual, API-driven UI content across 5+ languages; ensured WCAG 2.1 AA compliance and enterprise-grade NFRs.
- Recognized by TCS BANCS and TCS Interactive leadership for platform contributions.

Dibber Education Platform (Norway)

Jan 2024 – Sep 2024

React, Redux, Storybook

- Led frontend development for a back-office platform used across 600+ schools in 9 countries; built and maintained a Storybook-based component library of 40+ reusable components.
- Improved application performance by ~60% through lazy loading and render optimization; applied WCAG 2.0 accessibility standards.
- Received direct client appreciation and recognition from Global Head, TCS Interactive.

OMSA, Air India Express & Standard Bank – Enterprise Platforms

Feb 2022 – Nov 2024

React, TypeScript, REST APIs

- Delivered frontend solutions across insurance, airline booking and banking platforms; built 30+ reusable React components and API-driven, pixel-perfect responsive UI modules.

EDUCATION

M.Sc. in Electronic Science — Calcutta University, India (15 years full-time education) | Aug 2018 – Oct 2020

ACHIEVEMENTS

- Recognized by Global Head, TCS Interactive, for contributions to the Dibber platform (600+ schools, 9 countries).
- Awarded “Star of the Month” (2x) on the DSP project; appreciated by TCS BANCS and TCS Interactive leadership for enterprise-scale delivery across 5+ production applications serving 750K+ users.